
Chapter 5

Arithmetic Progression

Real-Life Story (Why Arithmetic Progression?)

Suppose your pocket money increases by 10 every week.

50, 60, 70, 80, ...

Here, the increase is always constant. Such patterns are called **Arithmetic Progressions (AP)**.

Arithmetic progressions appear in finance, engineering, economics, computer science, construction, and daily life calculations.

1 Arithmetic Progression

Overview

An **Arithmetic Progression (AP)** is a sequence of numbers in which the difference between consecutive terms remains constant.

We will learn:

- Common Difference
- Types of AP
- nth Term
- Sum of n Terms
- Arithmetic Mean
- Applications of AP

1.1 Definition

A sequence

$$a, a + d, a + 2d, a + 3d, \dots$$

is called an Arithmetic Progression if the difference between consecutive terms is constant.

$$d = a_2 - a_1$$

where:

$d = \text{Common Difference}$

Important Terms

- a is the **first term**
- d is the **common difference**
- a_n denotes the **nth term**
- S_n denotes the **sum of first n terms**

Remarks and Notation

- Consecutive terms differ by the same number.
- General notation:

$$a, a + d, a + 2d, \dots$$

- Example:

$$2, 5, 8, 11, \dots$$

$$d = 3$$

2 Types of Arithmetic Progression

1. Increasing AP

If

$$d > 0$$

then the AP increases.

Example:

$$3, 7, 11, 15, \dots$$

2. Decreasing AP

If

$$d < 0$$

then the AP decreases.

Example:

$$20, 17, 14, 11, \dots$$

3. Constant AP

If

$$d = 0$$

then all terms are equal.

Example:

$$5, 5, 5, 5, \dots$$

3 nth Term of AP

Key Formula

$$a_n = a + (n - 1)d$$

Example

Find the 10th term of:

$$2, 5, 8, 11, \dots$$

Given:

$$a = 2, \quad d = 3, \quad n = 10$$

Using formula:

$$a_n = a + (n - 1)d$$

$$a_{10} = 2 + (10 - 1)(3)$$

$$= 2 + 27$$

$$= 29$$

4 Sum of First n Terms

Formula:

$$S_n = \frac{n}{2}[2a + (n - 1)d]$$

Alternative formula:

$$S_n = \frac{n}{2}(a + l)$$

where l is the last term.

Example

Find the sum of first 20 terms:

$$3, 6, 9, 12, \dots$$

Given:

$$a = 3, \quad d = 3, \quad n = 20$$

$$S_n = \frac{20}{2} [2(3) + (20 - 1)(3)]$$

$$= 10[6 + 57]$$

$$= 630$$

5 Arithmetic Mean

If a, b, c are in AP, then:

$$b = \frac{a + c}{2}$$

Example

Find the arithmetic mean between 10 and 18.

$$AM = \frac{10 + 18}{2}$$

$$= 14$$

6 Visual Understanding



7 Applications of AP

- Salary increments
- Savings schemes
- Staircase designs
- Seating arrangements

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- Mobile recharge plans
 - Construction measurements

8 Solved Questions

Solved Question 1

Find the common difference of:

$$4, 9, 14, 19, \dots$$

Solution

$$d = 9 - 4$$

$$d = 5$$

Hence, the common difference is:

$$\boxed{5}$$

Solved Question 2

Find the 12th term of the AP:

$$3, 7, 11, 15, \dots$$

Solution

Given:

$$a = 3, \quad d = 4, \quad n = 12$$

Using:

$$a_n = a + (n - 1)d$$

$$a_{12} = 3 + (12 - 1)(4)$$

$$= 3 + 44$$

$$= 47$$

Hence,

$$\boxed{47}$$

Solved Question 3

Find the sum of first 15 terms of:

$$2, 5, 8, 11, \dots$$

Solution

Given:

$$a = 2, \quad d = 3, \quad n = 15$$

Using formula:

$$S_n = \frac{n}{2}[2a + (n-1)d]$$

$$S_{15} = \frac{15}{2}[2(2) + (15-1)(3)]$$

$$= \frac{15}{2}[4 + 42]$$

$$= \frac{15}{2}(46)$$

$$= 15 \times 23$$

$$= 345$$

Hence,

$$\boxed{345}$$

Solved Question 4

Find the arithmetic mean between 18 and 42.

Solution

$$AM = \frac{18 + 42}{2}$$

$$= \frac{60}{2}$$

$$= 30$$

Hence,

$$\boxed{30}$$

Solved Question 5

Determine whether the sequence

$$10, 6, 2, -2, \dots$$

is an AP.

Solution

$$6 - 10 = -4$$

$$2 - 6 = -4$$

$$-2 - 2 = -4$$

Since the common difference is constant, the sequence is an AP.

$$\boxed{d = -4}$$

Solved Question 6

Find the first term if:

$$a_8 = 31, \quad d = 4$$

Solution

Using:

$$a_n = a + (n - 1)d$$

$$31 = a + (8 - 1)(4)$$

$$31 = a + 28$$

$$a = 3$$

Hence,

$$\boxed{3}$$

Solved Question 7

Which term of the AP

$$5, 9, 13, \dots$$

is equal to 101?

Solution

Given:

$$a = 5, \quad d = 4$$

Using:

$$a_n = a + (n - 1)d$$

$$101 = 5 + (n - 1)(4)$$

$$101 = 5 + 4n - 4$$

$$101 = 4n + 1$$

$$100 = 4n$$

$$n = 25$$

Hence,

$25^{\text{th}} \text{ term}$

Solved Question 8

Find the sum of first 20 natural numbers.

Solution

Natural numbers form an AP:

$$1, 2, 3, \dots, 20$$

Given:

$$a = 1, \quad l = 20, \quad n = 20$$

Using:

$$S_n = \frac{n}{2}(a + l)$$

$$S_{20} = \frac{20}{2}(1 + 20)$$

$$= 10(21)$$

$$= 210$$

Hence,

210

Solved Question 9

Insert 3 arithmetic means between 4 and 20.

Solution

Sequence becomes:

$$4, a, b, c, 20$$

Total terms:

$$n = 5$$

Using:

$$d = \frac{l - a}{n - 1}$$

$$d = \frac{20 - 4}{4}$$

$$d = 4$$

Now add 4 repeatedly:

$$4, 8, 12, 16, 20$$

Hence, the arithmetic means are:

$8, 12, 16$

Solved Question 10

The sum of first n terms of an AP is:

$$S_n = 2n^2 + 3n$$

Find the n th term.

Solution

We know:

$$a_n = S_n - S_{n-1}$$

Given:

$$S_n = 2n^2 + 3n$$

First find:

$$S_{n-1} = 2(n-1)^2 + 3(n-1)$$

$$= 2(n^2 - 2n + 1) + 3n - 3$$

$$= 2n^2 - 4n + 2 + 3n - 3$$

$$= 2n^2 - n - 1$$

Now:

$$a_n = (2n^2 + 3n) - (2n^2 - n - 1)$$

$$= 2n^2 + 3n - 2n^2 + n + 1$$

$$= 4n + 1$$

Hence,

$$\boxed{a_n = 4n + 1}$$

9 Practice Questions

Basic Level

1. Find the common difference:

$$4, 9, 14, 19, \dots$$

2. Find the 15th term:

$$3, 7, 11, \dots$$

3. Find the sum of first 25 terms:

$$2, 5, 8, \dots$$

4. Find the arithmetic mean between 12 and 30.

5. Determine whether

$$10, 7, 4, 1, \dots$$

is an AP.

Intermediate Level

6. Find the 25th term:

$$8, 13, 18, \dots$$

7. Find:

$$1 + 4 + 7 + \dots + 100$$

8. Which term equals 101?

$$5, 9, 13, \dots$$

9. Find the first term if:

$$a_{12} = 50, d = 4$$

10. Find the sum of first 40 natural numbers.

Advanced Level

11. Insert 5 arithmetic means between:

6 and 30

12. If the 7th term is 35 and the 15th term is 67, find a and d .

13. If

$$S_n = 3n^2 + 2n$$

find the n th term.

14. Find the middle term:

5, 9, 13, ..., 101

15. How many terms of

7, 11, 15, ...

must be taken so that their sum is 468?

10 Quick Formula Sheet

Concept	Formula	Meaning
n th Term	$a_n = a + (n - 1)d$	General term
Sum of n Terms	$S_n = \frac{n}{2}[2a + (n - 1)d]$	Total sum
Arithmetic Mean	$AM = \frac{a + b}{2}$	Middle value

Practice Daily to Master Mathematics